No. of Functional Features Included in the Solution

SMARTBRIDGE

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Registration	Creates a new user account with encrypted bcrypt password hashing.	Auth Module (authController.js)	Done	0.5 Marks
2	User Login & JWT Auth	Verifies credentials and generates a signed JWT token.	Auth Module (authController.js)	Done	0.5 Marks
3	Get User Profile	Fetches profile details for the authenticated bearer token.	Auth Module (authController.js)	Done	0.5 Marks
4	Add Workout Record	Logs a workout with name, category, duration, calories, and date.	Workout Module (workoutController.js)	Done	0.5 Marks
5	Get / Search Workouts	Retrieves all workouts or filters history by workout name, category, or date.	Workout Module (workoutController.js)	Done	0.5 Marks
6	Update / Delete Workout	Modifies or deletes an existing workout	Workout Module (workoutController.js)	Done	0.5 Marks

		record linked to the user.			
7	AI Workout Plan Generator	Queries Gemini AI (@google/genai) using age, goals, and experience level.	AI Module (aiController.js, geminiService.js)	Done	1.0 Marks
8	AI Fitness Analytics Insights	Analyzes total workouts, avg duration, and calories burned to return insights.	AI Module (aiController.js, geminiService.js)	Done	1.0 Marks

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8 (100%)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	REST API JSON Consumer Interface / Postman / Thunder Client
2	Backend / Logic	3	Express Routing, Controller Business Logic, Error Handler Middleware
3	Database / Storage	2	MongoDB Atlas Users Collection, Workouts Collection
4	API / Integration	2	Google Gemini AI Recommendation & Fitness Insight Service Integration
5	Security / Authentication	2	JWT Bearer Authentication, bcryptjs Password Hashing